Globally optimal quadratic deficit bounds for two- and three-point reciprocal products

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Abstract

We give machine-checked proofs of quadratic error bounds with optimal global coefficients for the product of a sum and a reciprocal sum on a positive interval. In the two-point weighted problem, the optimal coefficient for squared Euclidean distance to the two maximizing endpoint pairs is $pq/(mM)$. In the unweighted three-point problem, the optimal coefficient for distance to the six endpoint triples with one or two upper endpoints is $2/(mM)$. The three-point proof uses six positive-polynomial certificates and an explicit covering of the ordered domain. A Lean development also verifies the classical exact fixed-weight partition formula as background. All formal theorems are proved without unfinished proof placeholders or additional axioms. The present note records verified statements and proofs; first-publication priority and editorial novelty are not claimed.

1 Setting and relation to classical results

Throughout, $0 < m < M$ and every input belongs to $[m, M]$. The classical Kantorovich inequality bounds

$$\left(\sum_i w_i x_i\right)\left(\sum_i w_i/x_i\right) \le \frac{(m+M)^2}{4mM}\left(\sum_i w_i\right)^2, \qquad w_i \ge 0.$$

Newman gives an elementary scalar reduction and convexity proof [1]. The exact fixed-weight maximum is also established literature: if $W = \sum_i w_i$ and $\delta^2 = \min_S(\sum_{i\in S} w_i - \sum_{i\notin S} w_i)^2$, then

$$Q_* = \frac{(m+M)^2W^2 - (M-m)^2\delta^2}{4mM}.$$

This formula appears in Niculescu–Persson, Section 1.4, Exercise 1 [2]. For equal weights, Lupaş gives the even/odd correction [3]. We use these results as historical context and supply independent formal proofs; they are not presented as new mathematics.

The question here is quantitative: if an input's product is close to its maximum, how close is that input to a maximizing endpoint configuration? The distance used below is the ordinary unweighted Euclidean distance on the coordinate space. Changing that metric changes the coefficient.

2 Two weighted points

For $p, q > 0$, put

$$Q(x,y) = (px+qy)(p/x+q/y), \qquad Q_* = (p+q)^2 + \frac{pq(M-m)^2}{mM}.$$

The maximizing set is $E_2 = \{(m, M), (M, m)\}$. Define

$$d_2^2(x, y) = \min\{(x - m)^2 + (y - M)^2, (x - M)^2 + (y - m)^2\}.$$

Theorem 1. *For every $x, y \in [m, M]$,*

$$Q_* - Q(x, y) \;\geq\; \frac{pq}{mM}\, d_2^2(x, y).$$

The coefficient $pq/(mM)$ is the largest coefficient valid for every input in the interval. Moreover, $Q(x, y) = Q_*$ if and only if $(x, y) \in E_2$.

Proof. Expansion gives the exact deficit identity

$$Q_* - Q = \frac{pq}{mM}\left((M - m)^2 - \frac{mM(x - y)^2}{xy}\right).$$

By symmetry assume $x \leq y$, so $d_2^2 = (x - m)^2 + (M - y)^2$. The numerator of the required scalar inequality has the following decomposition:

$$\begin{aligned} &xy\big((M - m)^2 - (x - m)^2 - (M - y)^2\big) - mM(y - x)^2 \\ &\quad = (x - m)(y - x)^2 y + 2xy(x - m)(y - x) \\ &\qquad + (M - y)(y - x)\big(2x^2 + (2x - m)(y - x)\big) \\ &\qquad + 2xy(x - m)(M - y). \end{aligned}$$

Every term is nonnegative. Divide by $xy > 0$ and use the deficit identity. At $x = y = m$, the distance equals $(M - m)^2 > 0$ and the deficit equals $pq(M - m)^2/(mM)$, proving optimality. The bound forces $d_2^2 = 0$ for a maximizer, and direct evaluation verifies both endpoint pairs. □

Corollary 1. *If $Q_* - Q(x, y) \leq \varepsilon$, then $d_2^2(x, y) \leq mM\varepsilon/(pq)$.*

The inequality itself permits $p = 0$ or $q = 0$ and $m = M$. The extremizer classification and the distance corollary above require positive weights. The formal optimal-coefficient statement includes zero weights, in which case the optimal coefficient is zero.

Optimality concerns the single coefficient valid on the entire interval. We do not assert that squared distance gives the optimal local rate near an endpoint maximizer; the distance corollary can be loose in that regime.

3 Three unweighted points

Put

$$Q(x, y, z) = (x + y + z)(1/x + 1/y + 1/z), \qquad Q_* = 9 + \frac{2(M - m)^2}{mM}.$$

Let E_3 consist of the six triples with coordinates in $\{m, M\}$ and exactly one or exactly two coordinates equal to M. Write $d_3^2 = \operatorname{dist}((x, y, z), E_3)^2$, explicitly the minimum of the six sums of squared coordinate differences.

Theorem 2. *For every $x, y, z \in [m, M]$,*

$$Q_* - Q(x, y, z) \;\geq\; \frac{2}{mM}\, d_3^2(x, y, z).$$

The coefficient $2/(mM)$ is the largest coefficient valid for every input in the interval.

3.1 Complete polynomial-certificate proof

Sort the inputs so that $m \le x \le y \le z \le M$. Define

$$N_L = xyz\big(2(M-m)^2 - 2((x-m)^2 + (y-m)^2 + (M-z)^2)\big) \\ - mM\big((x-y)^2 z + (x-z)^2 y + (y-z)^2 x\big).$$

Define N_U by replacing $(y-m)^2$ by $(y-M)^2$. Exact algebra gives

$$Q_* - Q - \frac{2}{mM}\big((x-m)^2 + (y-m)^2 + (M-z)^2\big) = \frac{N_L}{mMxyz},$$

and the analogous identity for N_U. Both denominators are positive. It is therefore sufficient to prove $N_L \ge 0$ when $2y \le m + M$ and $N_U \ge 0$ when $2y \ge m + M$.

Here is an exhaustive cover of the first region. In each row, all listed parameters are nonnegative; the columns give x, y, z, M respectively.

	x	y	z	M
0	$m + a$	$m + a + t$	$m + 2a + 2t + r$	$m + 2a + 2t + r + b$
1	$m + a$	$m + a + v + t$	$m + 2a + 2v + t$	$m + 2a + 2v + 2t + r$
2	$m + u + a$	$m + u + a + t$	$m + 2u + a + t$	$m + 2u + 2a + 2t + r$

To verify coverage, first test $2y \le z + m$. If true, use row 0 with $(a, t, r, b) = (x - m, y - x, z - 2y + m, M - z)$. Otherwise test $x - m \le z - y$. If true, use row 1 with

$$(a, v, t, r) = (x - m, z - y - (x - m), 2y - z - m, M + m - 2y).$$

If false, use row 2 with

$$(u, a, t, r) = (z - y, x - m - (z - y), y - x, M + m - 2y).$$

The interval, ordering, and half-region inequalities prove every parameter nonnegative. Substitution reproduces the original four coordinates exactly.

Substituting these three rows into N_L gives positive-coefficient polynomials with respectively 66, 71, and 83 monomials. Their complete integer-coefficient expressions are supplied in the formal supplement. For the upper half, reflect the ordered triple by

$$(x, y, z) \longmapsto (m + M - z, m + M - y, m + M - x).$$

Substituting the same three rows into the reflected N_U gives positive-coefficient polynomials with 89, 107, and 115 monomials. Reflection maps the upper half to the lower half, so this also exhausts the upper region. Each of the six expansion identities and its positivity is checked separately in Lean.

The squared distance d_3^2 is no larger than the corresponding squared distance used in N_L or N_U. This proves the theorem for sorted inputs. The product and the six-point distance are permutation invariant; the formal proof checks all six orderings, completing the unrestricted case.

Finally, at $x = y = z = m$, the distance is $(M - m)^2$ and the deficit is $2(M - m)^2/(mM)$. Thus a larger coefficient is impossible. Endpoint evaluation gives $Q(m, m, M) = Q_*$, confirming the asserted maximum.

4 Verification and scope

The accompanying source pins Lean 4.34.0 and mathlib commit

`5ed2965256430c3649e86755f9576b54eca72435`

The formal source proves Theorems 1 and 2, characterizes their best coefficients, and gives the two-point distance certificate. The polynomial generator helps discover the certificates. The formal checker verifies the resulting identities and their signs; it does not trust an external numerical oracle.

Every declared result in the verification record has a complete proof. Axiom audits permit only the standard Lean logical axioms `propext`, `Classical.choice`, and `Quot.sound`. Targeted kernel replay checks the saved declarations. This is not a claim of verification in a second independent proof assistant, physical-system validation, or floating-point numerical error certification.

Exact finite computations suggest a dimension-dependent extension for larger unweighted systems. That extension is not proved here and is not part of the publication's verified claims.

4.1 Exact examples

For $m = 1, M = 4$ and unit weights, the following calculations are also checked in the formal source. A stability equality can have positive distance, so it must not be confused with an objective maximizer.

Problem	Input	Squared distance	Deficit
Two points	$(1, 1)$	9	9/4
Three points	$(1, 1, 1)$	9	9/2
Three points	$(1, 5/2, 4)$	9/4	9/8

5 Prior-work and publication status

The fixed-weight partition formula and the equal-weight parity correction are explicitly attributed above. A recent weighted Schweitzer refinement [4] is relevant neighboring work; its accessible HTML omits the equations and its full remainder could not be compared. Consequently, absence of an exact search match is not evidence of priority for our distance estimates. The contribution of this release is a reproducible, fully checked development with explicit globally optimal coefficients and finite certificates. Editorial acceptance and first-publication novelty have not been established.

References

[1] M. Newman, *Kantorovich's inequality*, Journal of Research of the National Bureau of Standards B 64B (1960), 33–34. Original paper (NIST PDF).

[2] C. P. Niculescu and L.-E. Persson, *Convex Functions and Their Applications*, manuscript dated 16 September 2004, Section 1.4, Exercise 1, p. 32. Original manuscript (PDF).

[3] A. Lupaş, *A remark on the Schweitzer and Kantorovich inequalities*, Publikacije Elektrotehničkog Fakulteta, Serija Matematika i Fizika (1972), 13–15. Source PDF.

[4] Alqahtani and M. Z. Sarikaya, *Refined Weighted Schweitzer-Type Inequalities With Applications to Statistical Moments*, Journal of Mathematics (2026), article 4822611. doi:10.1155/jom/4822611.